3.2 Light - Mark scheme

Cambridge IGCSE Physics (0625) | Syllabus 2026–2028

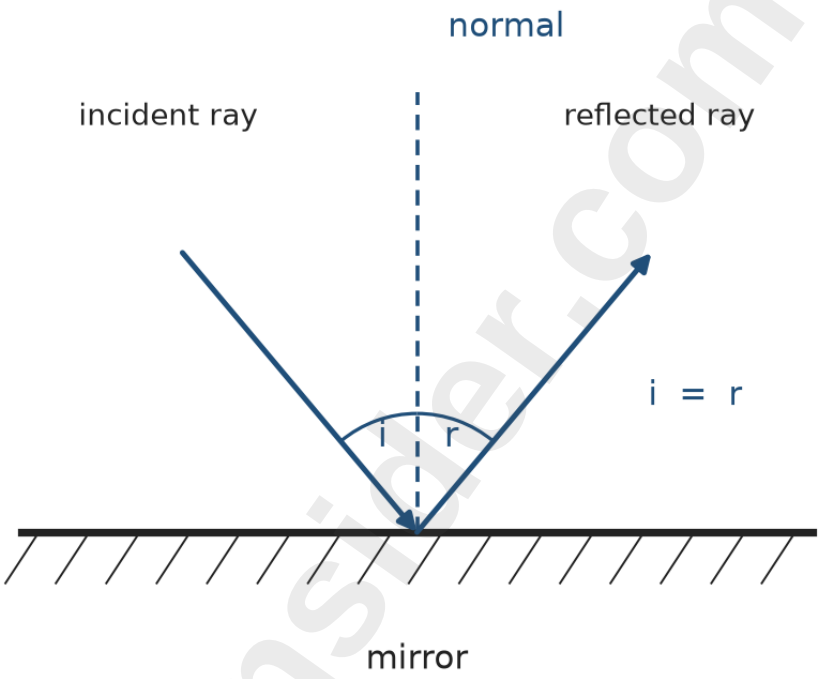
Covers: 3.2.1 Reflection | 3.2.2 Refraction | 3.2.3 Thin lenses | 3.2.4 Dispersion

Tier: Core and Extended

For teachers. Contains the answers to Sections 6 and 7 and model versions of the Section 4 diagrams.

Model diagrams (Section 4)

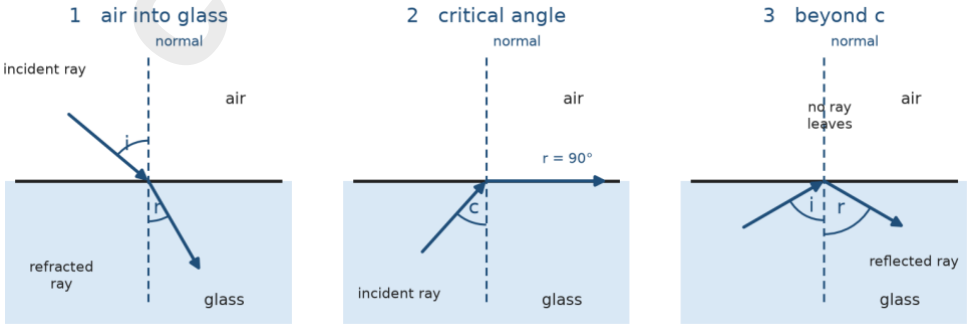
4.1 Reflection at a plane mirror Core



Both angles are measured from the normal. The law of reflection is i = r. In the model the angles are 40°.

Credit for: Arrow on the incident ray, towards the mirror; Arrow on the reflected ray, away from the mirror; i between the incident ray and the normal; r between the reflected ray and the normal; i equal to r.

4.2 Refraction and total internal reflection Core

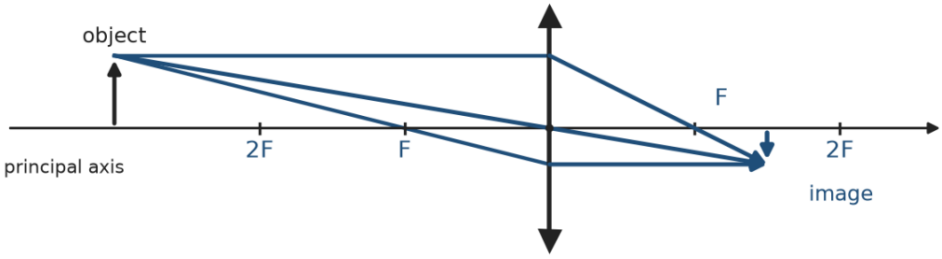


Sketch 1 uses i = 50° and r = 30°. Sketch 2 uses c = 42°, so the angle of refraction is 90°. Sketch 3 uses 60°, which is greater than 42°, so the ray is totally internally reflected and i = r.

Credit for: Sketch 1: ray bends towards the normal as it enters the glass; i and r both measured from the normal, with r smaller than i; Sketch 2: refracted ray along the boundary and c labelled; Sketch 3: reflected ray only, with the angle of reflection equal to the angle of incidence; Arrows showing the direction of each ray.

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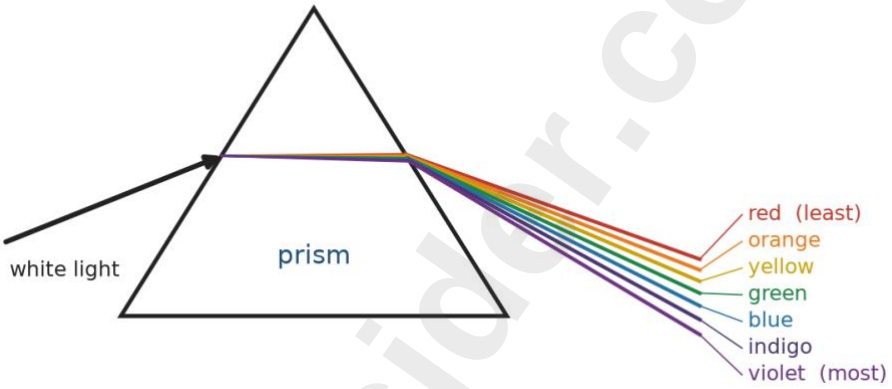
4.3 Real image formed by a converging lens Core



A third ray, through F on the object side, leaves the lens parallel to the principal axis and passes through the same image point. The image is real, inverted and diminished.

Credit for: One ray parallel to the principal axis, refracted through F on the far side; One ray through the optical centre, not bent; Image arrow where the two rays meet; Image inverted, between F and 2F on the opposite side to the object; Image shorter than the object.

4.4 Dispersion of white light by a prism Core



The spreading is dispersion. A rectangular block shifts the beam sideways but does not spread it into a spectrum, and the ray that leaves is parallel to the ray that entered.

Credit for: White light labelled on the way in; The beam changes direction at each face; Seven colours in order: red, orange, yellow, green, blue, indigo, violet; Red labelled as refracted least (longest wavelength, lowest frequency); Violet labelled as refracted most (shortest wavelength, highest frequency).

Multiple-choice answers (Section 6)

Each wrong option is a specific mistake. Where an examiner report records candidates making it, the series and paper are given in brackets. Use the third column to diagnose, not just to mark.

#	Answer	Why each wrong option is wrong
1	**A**	**B** Gives the angle between the incident ray and the reflected ray, 2 × 28° = 56°. The angle of incidence is between the incident ray and the normal. **C** Uses 62°, the angle between the ray and the mirror. The angle of incidence is measured from the normal, so it is 90° − 62° = 28°. (June 2026, Paper 23) **D** 180° − 62° = 118°, the angle between the ray and the other part of the mirror. Neither angle is measured from the normal.

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#	Answer	Why each wrong option is wrong
2	**D**	**A** Puts the image in the plane of the mirror. The image is as far behind the mirror as the object is in front. (Nov 2022, Paper 13) **B** The distance is right, but a plane-mirror image is virtual. The reflected rays do not meet behind the mirror. (June 2023, Paper 13) **C** Doubles the object distance. The image distance equals the object distance, so it is 12 cm, not 24 cm.
3	**B**	**A** Leaves a refracted ray in the air. Above the critical angle, no light leaves the glass. (June 2024, Paper 11) **C** Reverses the media. Total internal reflection is for a ray already in the glass (or water) meeting air, not for a ray in air meeting glass. (March 2025, Paper 22) **D** A ray crosses without a change of direction only when it meets the boundary along the normal, so the angle of incidence is 0°.
4	**A**	**B** Reverses the frequencies. Red light has a lower frequency than violet light, and a longer wavelength. (Nov 2023, Paper 12) **C** Reverses the wavelengths. Red has the longer wavelength; violet has the shorter one. **D** The seven colours have different frequencies. That is why a prism disperses them.
5	**C**	**A** The speed does fall, but entering the slower medium bends the ray towards the normal, not away from it. **B** Glass is optically denser than air, so the light slows down. It does not speed up. **D** Refraction at an angle happens because the speed changes. The frequency stays the same; the speed and the wavelength do not.
6	**B**	**A** Divides the wrong way up: sin 30° / sin 50° = 0.653. The refractive index is sin i / sin r. **C** Divides the angles, 50 / 30 = 1.67, and does not use the sines. **D** Works out 1 / sin 30° = 2.00. That uses the critical-angle equation on the angle of refraction.
7	**C**	**A** Gives sin 34° = 0.559 and stops. n = 1 / sin c, so sin c belongs in the denominator: 1 / 0.559 = 1.79. (June 2023, Paper 43) **B** Uses the complement, 90° − 34° = 56°, in 1 / sin 56° = 1.21. The critical angle is already measured from the normal. **D** Divides the angles, 90 / 34 = 2.65, and does not use the sine.
8	**D**	**A** The image is real, inverted and diminished. A magnifying glass gives a virtual, upright, enlarged image. **B** Rays that leave the lens are parallel, so they do not form an image at a finite distance. **C** Selects an enlarged image that is real. The image in a magnifying glass is enlarged and virtual. (June 2023, Paper 23)

Mark scheme (Section 7)

Question 1 [5]

(a) Angle of reflection = 35° **[1]**

(b) Angle with the mirror = 90° − 35° = 55° **[1]**

55° is the angle between the ray and the mirror. It is not the angle of incidence.

(c) Draw two rays from the object point to the mirror **[1]**

For each ray, draw the reflected ray with the angle of reflection equal to the angle of incidence (both from the normal) **[1]**

Extend the two reflected rays back behind the mirror; they meet at the image **[1]**

The perpendicular construction is an alternative full answer: extend the perpendicular from the object through the mirror, and mark the image the same distance behind the mirror as the object is in front.

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Question 2 [3]

Same size as the object **[1]**

8.0 cm behind the mirror (the same distance from the mirror as the object) **[1]**

Virtual / cannot be formed on a screen **[1]**

Laterally inverted is a correct extra point. It does not replace 'virtual'.

Question 3 [6]

(a) A narrow ray from a ray box, or a laser, is aimed at a transparent block, and the outline of the block is drawn **[1]**

The incident ray and the ray that leaves are marked, and joined to show the path inside the block **[1]**

The experiment is repeated with a block of a different shape, for example a semicircular block or a prism **[1]**

Do not accept diffraction in place of refraction. The June 2021, Paper 11 report records candidates who confused the two.

(b) On entering the glass the light slows down and bends towards the normal, so the angle of refraction is smaller than the angle of incidence **[1]**

On leaving the glass the light speeds up and bends away from the normal **[1]**

For this rectangular block the ray that leaves is parallel to the ray that entered (it is shifted sideways) **[1]**

Question 4 [5]

(a) The angle of incidence in the denser medium (the glass) **[1]**

for which the angle of refraction is 90° (the refracted ray travels along the boundary) **[1]**

Do not accept a statement of the condition for total internal reflection (angle of incidence greater than the critical angle) as the meaning of the critical angle itself.

(b) Less than the critical angle: some light is refracted out into the air and some is reflected back into the glass **[1]**

Greater than the critical angle: total internal reflection. All of the light is reflected and no ray leaves the glass **[1]**

At exactly the critical angle the refracted ray is along the boundary. That is not total internal reflection.

(c) One of: optical fibre, reflecting prism in binoculars, cat's eye / bicycle reflector, endoscope **[1]**

Question 5 [5]

(a) The white light is split into a spectrum of colours **[1]**

The effect is dispersion **[1]**

Do not accept refraction or diffraction as the name of the spreading. The June 2024, Paper 31 report records both of those errors.

The outgoing coloured rays are not parallel to the incoming beam. That pattern belongs to a rectangular block. The November 2022, Paper 32 report records candidates who drew the prism that way.

(b) red, orange, yellow, green, blue, indigo, violet, and no other order **[1]**

Red has the lower frequency and the longer wavelength. The November 2023, Paper 12 report records candidates who learned the colour order by rote without recognising that red light has a lower frequency and a higher wavelength than violet light.

(c) Red has the longest wavelength **[1]**

Violet is refracted the most **[1]**

Question 6 [4] Extended

(a) $\sin c = 1 / 1.6 = 0.625$ **[1]**

$c = \sin^{-1}(0.625) = 38.7°$ **[1]**

Not $c = 1 / \sin n$. The March 2023, Paper 42 report records that rearrangement. Here $\sin c = 1 / n$, then c is the inverse sine.

(b) 30° is less than 38.7°, so total internal reflection does not occur **[1]**

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A refracted ray leaves the glass. Some light is also reflected inside the glass **[1]**

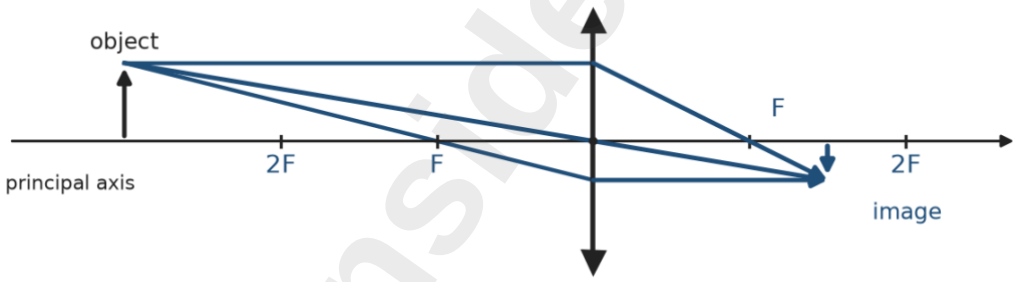
Question 7 [3] Extended

At each bounce inside the fibre, the angle of incidence is greater than the critical angle **[1]**
The light is totally internally reflected, so it stays in the fibre **[1]**
One telecommunications use: carrying telephone signals, internet signals or other data over a long distance **[1]**
The word reflection on its own is not enough. The November 2021, Paper 42 report gave no credit for stating only reflection.

Question 8 [4]

(a) The rays converge and meet at the principal focus on the other side of the lens **[1]**
(b) The rays diverge, and appear to have come from the principal focus on the same side as the incident light **[1]**
(c) The distance from the centre (optical centre) of the lens to the principal focus **[1]**
Not the distance between the two principal focuses, and not the distance from the object to the lens.
(d) The point on the principal axis where rays parallel to the principal axis meet after passing through the lens **[1]**
Not the centre of the lens. The June 2026, Paper 32 report records candidates who put the principal focus there.
Do not shorten principal focus to focus. The June 2022, Paper 41 report says focus cannot be used as a shortened form of principal focus.

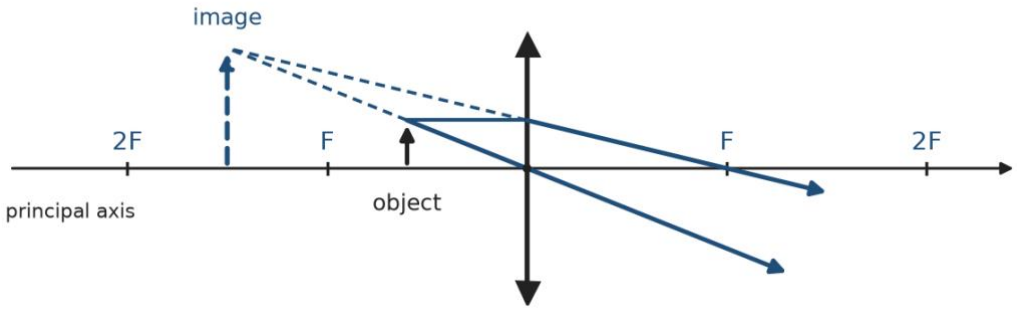
Question 9 [5]



(a) Ray parallel to the principal axis, refracted through F on the far side **[1]**
Ray through the optical centre, continuing in a straight line **[1]**
Image arrow drawn inverted where the two rays meet, between F and 2F on the far side, and shorter than the object **[1]**
The third standard ray (through the near F, then out parallel to the axis) may replace either of the two rays above.
Do not bend the ray through the optical centre down the principal axis.
(b) Real and inverted **[1]**
Diminished **[1]**
Do not award the description mark if the answer pairs contradictory words, such as upright and inverted, or real and virtual.

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Question 10 [7] Extended



(a) Ray parallel to the principal axis, refracted towards the far F, then extrapolated back with a dotted line **[1]**

Ray through the optical centre in a straight line, extrapolated back with a dotted line **[1]**

Upright image, taller than the object, on the same side of the lens as the object, where the dotted lines meet **[1]**

Say 'on the same side of the lens as the object'. The November 2022, Paper 42 report did not credit the vaguer phrase 'in front of the lens'.

A solid image on the far side of the lens is the wrong construction: that is a real image.

(b) Virtual, upright and enlarged **[1]**

Magnifying glass **[1]**

(c) Long-sightedness: a converging lens **[1]**

Short-sightedness: a diverging lens **[1]**

Long-sight means near objects are not focused on the retina and distant objects are clearer. It does not mean that distant objects are blurred.

Question 11 [3] Extended

(a) Yes, because monochromatic light is light of a single frequency (a single wavelength is an alternative wording) **[1]**

Do not accept 'one colour' on its own. One colour can cover a range of frequencies.

(b) speed in the glass = speed in air / n = 3.0×10^8 / 1.50 **[1]**

= 2.0×10^8 m/s **[1]**

Not $3.0 \times 10^8 \times 1.50 = 4.5 \times 10^8$ m/s. The June 2024, Paper 41 report records candidates who multiplied the speed of light in air by n.

Syllabus coverage

Where each checklist statement is practised in this booklet.

#	Statement	Practised in
1	Define and use the terms normal, angle of incidence and angle of reflection	Diagram 4.1, Q1(a), (b), MCQ1
2	Describe the formation of an optical image by a plane mirror and give its characteristics, i.e. same size, same distance from mirror, virtual	WE3, Q2, MCQ2
3	State that for reflection, the angle of incidence is equal to the angle of reflection; recall and use this relationship	Diagram 4.1, Q1(a)
4	Use simple constructions, measurements and calculations for reflection by plane mirrors	Q1(c)
5	Define and use the terms normal, angle of incidence and angle of refraction	Diagram 4.2, Q3(b), MCQ5
6	Describe an experiment to show refraction of light by transparent blocks of different shapes	Q3(a)

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#	Statement	Practised in
7	Describe the passage of light through a transparent material (limited to the boundaries between two mediums only)	Q3(b), Diagram 4.2
8	State the meaning of critical angle	Q4(a), WE2, MCQ3
9	Describe internal reflection and total internal reflection using both experimental and everyday examples	Diagram 4.2, Q4(b), (c), Q7, MCQ3
10	Define refractive index, n, as the ratio of the speeds of a wave in two different regions	Q11(b)
11	Recall and use the equation n = sin i / sin r	WE1, MCQ6
12	Recall and use the equation n = 1 / sin c	WE2, Q6(a), MCQ7
13	Describe the use of optical fibres, particularly in telecommunications	Q7
14	Describe the action of thin converging and thin diverging lenses on a parallel beam of light	Q8(a), (b), Diagram 4.3
15	Define and use the terms focal length, principal axis and principal focus (focal point)	Q8(c), (d), Diagram 4.3, Q9
16	Draw and use ray diagrams for the formation of a real image by a converging lens	Diagram 4.3, Q9
17	Describe the characteristics of an image using the terms enlarged/same size/diminished, upright/inverted and real/virtual	Q2, Q9(b), Q10(b), MCQ2
18	Know that a virtual image is formed when diverging rays are extrapolated backwards and does not form a visible projection on a screen	WE3, Q2, Q10(a), MCQ2
19	Draw and use ray diagrams for the formation of a virtual image by a converging lens	Q10(a)
20	Describe the use of a single lens as a magnifying glass	Q10(b), MCQ8
21	Describe the use of converging and diverging lenses to correct long-sightedness and short-sightedness	Q10(c)
22	Describe the dispersion of light as illustrated by the refraction of white light by a glass prism	Diagram 4.4, Q5(a)
23	Know the traditional seven colours of the visible spectrum in order of frequency and in order of wavelength	Diagram 4.4, Q5(b), (c), MCQ4
24	Recall that visible light of a single frequency is described as monochromatic	Q11(a)

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